

**LUNA2000B**  
**ESS**

# **Modbus Port Definitions**

|              |            |
|--------------|------------|
| <b>Issue</b> | 01         |
| <b>Date</b>  | 2025-09-10 |



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## **Huawei Technologies Co., Ltd.**

Address: Huawei Industrial Base  
Bantian, Longgang  
Shenzhen 518129  
People's Republic of China

Website: <https://e.huawei.com>

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# 1 ESS Models

This chapter describes the energy storage system (ESS) models that use Modbus protocol and the minimum software versions required.

1.1 ESS Model Description

## 1.1 ESS Model Description

Table 1-1 ESS models and software versions

| Model Series           | Model Name                                                                                            | Minimum Firmware Version    |
|------------------------|-------------------------------------------------------------------------------------------------------|-----------------------------|
| LUNA2000-215KWH series | LUNA2000-161-2S11<br>LUNA2000-107-1S11<br>LUNA2000-215-2S10<br>LUNA2000-215-2S12<br>LUNA2000-215-2S11 | LUNA2000B V200R024C00SPC310 |

 NOTE

Different ESS models vary in the number of ESS subsystems and configuration of commercial & industrial (C&I) cabinet subsystems. Check the information in the point table before use.

# 2 Introduction

Modbus is a widely used protocol for device communications. This document describes the Modbus protocol used by Huawei ESSs and can be used to regulate subsequent development for third-party integration. Huawei ESSs comply with the standard Modbus protocol, and this document focuses on the information specific to Huawei ESSs. For other information about Modbus, see the standard documents about the Modbus protocol. For details about the interaction modes and examples of the standard protocol and custom part used in Huawei ESSs, see [4 Overview of the Communications Protocol](#)

## 2.1 Terms and Abbreviations

## 2.1 Terms and Abbreviations

Table 2-1 Terms and abbreviations

| Term              | Description                                                                                                             |
|-------------------|-------------------------------------------------------------------------------------------------------------------------|
| Master node       | In the master-slave communication, the party that initiates the communication is called the master node.                |
| Slave node        | During master-slave communication, the party that responds to a communication request is referred to as the slave node. |
| Broadcast address | Fixed to 0.                                                                                                             |
| Register address  | Recorded in two bytes.                                                                                                  |
| U16               | 16-bit unsigned integer                                                                                                 |
| U32               | 32-bit unsigned integer                                                                                                 |
| U64               | 64-bit unsigned integer                                                                                                 |
| I16               | 16-bit signed integer                                                                                                   |
| I32               | 32-bit signed integer                                                                                                   |

| Term       | Description                                 |
|------------|---------------------------------------------|
| I64        | 64-bit signed integer                       |
| STR        | Character string                            |
| MLD        | Multiple bytes                              |
| Bitfield16 | 16-bit data                                 |
| Bitfield32 | 32-bit data                                 |
| -          | N/A                                         |
| s          | Second                                      |
| EPOCHTIME  | Number of seconds since 1970-01-01 00:00:00 |
| RO         | Read-only data                              |
| RW         | Data that is readable and writable          |
| WO         | Write-only data                             |

# 3 C&I Cabinet Subsystem

- 3.1 Register Definitions
- 3.2 Alarm Definitions

## 3.1 Register Definitions

Table 3-1 C&I cabinet subsystem

| No. | Signal                                  | Re<br>ad/<br>Wr<br>ite | Ty<br>pe | Un<br>it | Gai<br>n | Addr<br>ess | Qua<br>ntit<br>y | Scope        |
|-----|-----------------------------------------|------------------------|----------|----------|----------|-------------|------------------|--------------|
| 1   | Local time                              | RO                     | U3<br>2  | N/<br>A  | 1        | 3016<br>5   | 2                | Epoch second |
| 2   | Rated capacity                          | RO                     | U3<br>2  | kW<br>h  | 10<br>00 | 3023<br>6   | 2                |              |
| 3   | Rated power                             | RO                     | U3<br>2  | kW       | 10<br>00 | 3023<br>8   | 2                |              |
| 4   | SOC                                     | RO                     | U1<br>6  | %        | 1        | 3040<br>0   | 1                | [0, 100]     |
| 5   | Charge/<br>Discharge<br>power           | RO                     | I32      | kW       | 10<br>00 | 3041<br>7   | 2                |              |
| 6   | Chargeable<br>capacity                  | RO                     | U3<br>2  | kW<br>h  | 10<br>00 | 3041<br>9   | 2                |              |
| 7   | Dischargeable<br>capacity               | RO                     | U3<br>2  | kW<br>h  | 10<br>00 | 3042<br>1   | 2                |              |
| 8   | Total auxiliary<br>power<br>consumption | RO                     | U3<br>2  | kW<br>h  | 10<br>0  | 3045<br>3   | 2                |              |



| No. | Signal                       | Read/Write | Type           | Unit | Gain | Address | Quantity | Scope                                                        |
|-----|------------------------------|------------|----------------|------|------|---------|----------|--------------------------------------------------------------|
| 9   | [Teleindication]<br>Alarm 1  | RO         | Bitfield<br>16 | N/A  | 1    | 30455   | 1        | For details about the bit definition, see alarm definitions. |
| 10  | [Teleindication]<br>Alarm 2  | RO         | Bitfield<br>16 | N/A  | 1    | 30456   | 1        | For details about the bit definition, see alarm definitions. |
| 11  | [Teleindication]<br>Alarm 3  | RO         | Bitfield<br>16 | N/A  | 1    | 30457   | 1        | For details about the bit definition, see alarm definitions. |
| 12  | [Teleindication]<br>Alarm 4  | RO         | Bitfield<br>16 | N/A  | 1    | 30458   | 1        | For details about the bit definition, see alarm definitions. |
| 13  | [Teleindication]<br>Alarm 5  | RO         | Bitfield<br>16 | N/A  | 1    | 30459   | 1        | For details about the bit definition, see alarm definitions. |
| 14  | [Teleindication]<br>Alarm 6  | RO         | Bitfield<br>16 | N/A  | 1    | 30460   | 1        | For details about the bit definition, see alarm definitions. |
| 15  | [Teleindication]<br>Alarm 7  | RO         | Bitfield<br>16 | N/A  | 1    | 30461   | 1        | For details about the bit definition, see alarm definitions. |
| 16  | [Teleindication]<br>Alarm 8  | RO         | Bitfield<br>16 | N/A  | 1    | 30462   | 1        | For details about the bit definition, see alarm definitions. |
| 17  | [Teleindication]<br>Alarm 9  | RO         | Bitfield<br>16 | N/A  | 1    | 30463   | 1        | For details about the bit definition, see alarm definitions. |
| 18  | [Teleindication]<br>Alarm 10 | RO         | Bitfield<br>16 | N/A  | 1    | 30464   | 1        | For details about the bit definition, see alarm definitions. |

| No. | Signal                                   | Read/Write | Type        | Unit  | Gain | Address | Quantity | Scope                                                        |
|-----|------------------------------------------|------------|-------------|-------|------|---------|----------|--------------------------------------------------------------|
| 19  | [Teleindication] Alarm 11                | RO         | Bitfield 16 | N/A   | 1    | 30465   | 1        | For details about the bit definition, see alarm definitions. |
| 20  | [Teleindication] Alarm 12                | RO         | Bitfield 16 | N/A   | 1    | 30466   | 1        | For details about the bit definition, see alarm definitions. |
| 21  | Active power of auxiliary power supply   | RO         | I32         | kW    | 1000 | 30497   | 2        |                                                              |
| 22  | Reactive power of auxiliary power supply | RO         | I32         | kV ar | 1000 | 30499   | 2        |                                                              |
| 23  | Power factor of auxiliary power supply   | RO         | I16         | N/A   | 1000 | 30501   | 1        |                                                              |
| 24  | [Teleindication] Alarm 13                | RO         | Bitfield 16 | N/A   | 1    | 30510   | 1        | For details about the bit definition, see alarm definitions. |
| 25  | [Teleindication] Alarm 14                | RO         | Bitfield 16 | N/A   | 1    | 30511   | 1        | For details about the bit definition, see alarm definitions. |
| 26  | [Teleindication] Alarm 15                | RO         | Bitfield 16 | N/A   | 1    | 30512   | 1        | For details about the bit definition, see alarm definitions. |
| 27  | [Teleindication] Alarm 16                | RO         | Bitfield 16 | N/A   | 1    | 30513   | 1        | For details about the bit definition, see alarm definitions. |
| 28  | [Teleindication] Alarm 17                | RO         | Bitfield 16 | N/A   | 1    | 30514   | 1        | For details about the bit definition, see alarm definitions. |

| No. | Signal                                | Read/Write | Type        | Unit | Gain | Address | Quantity | Scope                                                           |
|-----|---------------------------------------|------------|-------------|------|------|---------|----------|-----------------------------------------------------------------|
| 29  | [Teleindication] Alarm 18             | RO         | Bitfield 16 | N/A  | 1    | 30515   | 1        | For details about the bit definition, see alarm definitions.    |
| 30  | [Teleindication] Alarm 19             | RO         | Bitfield 16 | N/A  | 1    | 30516   | 1        | For details about the bit definition, see alarm definitions.    |
| 31  | [Teleindication] Alarm 20             | RO         | Bitfield 16 | N/A  | 1    | 30517   | 1        | For details about the bit definition, see alarm definitions.    |
| 32  | Battery cabin temperature 1           | RO         | I16         | °C   | 10   | 30703   | 1        |                                                                 |
| 33  | Battery cabin humidity 1              | RO         | I16         | %    | 10   | 30704   | 1        |                                                                 |
| 34  | Battery cabin temperature 2           | RO         | I16         | °C   | 10   | 30705   | 1        |                                                                 |
| 35  | Battery cabin humidity 2              | RO         | I16         | %    | 10   | 30706   | 1        |                                                                 |
| 36  | CO concentration 1                    | RO         | U16         | ppm  | 1    | 30715   | 1        |                                                                 |
| 37  | Exhaust fan speed 1                   | RO         | U16         | RP M | 1    | 30735   | 1        |                                                                 |
| 38  | Exhaust fan speed 2                   | RO         | U16         | RP M | 1    | 30736   | 1        | N/A                                                             |
| 39  | LTMS Working status                   | RO         | U16         | N/A  | 1    | 31400   | 1        | 0: Off<br>1: Self-circulating<br>2: Refrigeration<br>3: Heating |
| 40  | Outdoor temperature                   | RO         | I16         | °C   | 10   | 31411   | 1        | [-50.0, 150.0]                                                  |
| 41  | Battery-side supply water temperature | RO         | I16         | °C   | 10   | 31473   | 1        | [-50.0, 150.0]                                                  |

| No. | Signal                                      | Re<br>ad/<br>Wr<br>ite | Ty<br>pe | Un<br>it | Gai<br>n | Addr<br>ess | Qua<br>ntit<br>y | Scope          |
|-----|---------------------------------------------|------------------------|----------|----------|----------|-------------|------------------|----------------|
| 42  | Battery-side<br>return water<br>temperature | RO                     | I16      | °C       | 10       | 3147<br>4   | 1                | [-50.0, 150.0] |
| 43  | Power-side<br>supply water<br>temperature   | RO                     | I16      | °C       | 10       | 3147<br>5   | 1                | [-50.0, 150.0] |
| 44  | Power-side<br>return water<br>temperature   | RO                     | I16      | °C       | 10       | 3147<br>6   | 1                | [-50.0, 150.0] |

| No. | Signal       | Re<br>ad/<br>Wr<br>ite | Typ<br>e | Un<br>it | Gai<br>n | Addr<br>ess | Qua<br>ntit<br>y | Scope                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
|-----|--------------|------------------------|----------|----------|----------|-------------|------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 45  | Work status  | RO                     | U1<br>6  | N/<br>A  | 1        | 3203<br>4   | 1                | 0x0000 Idle<br>0x0100 Launch<br>0x0200 Running :<br>PQ Running<br>0x0201 Running :<br>VSG Running<br>0x0202 Running :<br>Hot Spare<br>0x0204 Running :<br>Limited power<br>0x0205 Running :<br>Power module<br>derating<br>0x0206 Running :<br>Battery current<br>limit<br>0x0207 Running :<br>Energy storage<br>system derating<br>0x0208 Running :<br>Laching<br>0x0300 Abnormal<br>shutdown<br>0x0301 Command<br>shutdown<br>0x0302 Power-on<br>authorization is<br>not enabled.<br>0x0303 Power Off<br>0x0304 EPO Off<br>0x0305 AC Off<br>0x0400 Ready for<br>spot check<br>0x0401 Spot check<br>in progress<br>0xB000 Offline<br>0xC000 Loading |
| 46  | Rack voltage | RO                     | l16      | V        | 10       | 3203<br>6   | 1                | N/A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

| No. | Signal                      | Read/Write | Type        | Unit | Gain | Address | Quantity | Scope                                                               |
|-----|-----------------------------|------------|-------------|------|------|---------|----------|---------------------------------------------------------------------|
| 47  | Rack current                | RO         | I16         | A    | 10   | 32037   | 1        | N/A                                                                 |
| 48  | SOH                         | RO         | U16         | %    | 1    | 32041   | 1        | [0,100]                                                             |
| 49  | SOE                         | RO         | U16         | %    | 1    | 32042   | 1        | [0,100]                                                             |
| 50  | DOD                         | RO         | U16         | %    | 1    | 32043   | 1        | [0,100]                                                             |
| 51  | Control SOC                 | RO         | U16         | %    | 10   | 32044   | 1        | [0.0,100.0]                                                         |
| 52  | Charging status             | RO         | U16         | N/A  | 1    | 32046   | 1        | 0: Idle<br>1: Recharge request<br>2: Recharging<br>3: Charging ends |
| 53  | Highest pack temperature    | RO         | I16         | °C   | 100  | 32066   | 1        | N/A                                                                 |
| 54  | Pack of highest temperature | RO         | U16         | N/A  | 1    | 32067   | 1        | N/A                                                                 |
| 55  | Lowest pack temperature     | RO         | I16         | °C   | 100  | 32068   | 1        | N/A                                                                 |
| 56  | Pack of lowest temperature  | RO         | U16         | N/A  | 1    | 32069   | 1        | N/A                                                                 |
| 57  | Lowest pack voltage         | RO         | U16         | V    | 10   | 32070   | 1        | N/A                                                                 |
| 58  | Pack of lowest voltage      | RO         | U16         | N/A  | 1    | 32071   | 1        | N/A                                                                 |
| 59  | Highest pack voltage        | RO         | U16         | V    | 10   | 32072   | 1        | N/A                                                                 |
| 60  | Pack of highest voltage     | RO         | U16         | N/A  | 1    | 32073   | 1        | N/A                                                                 |
| 61  | [Teleindication] Alarm 21   | RO         | Bitfield 16 | N/A  | 1    | 32133   | 1        | For details about the bit definition, see alarm definitions.        |

| No. | Signal                       | Read/Write | Type           | Unit | Gain | Address   | Quantity | Scope                                                        |
|-----|------------------------------|------------|----------------|------|------|-----------|----------|--------------------------------------------------------------|
| 62  | [Teleindication]<br>Alarm 22 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>4 | 1        | For details about the bit definition, see alarm definitions. |
| 63  | [Teleindication]<br>Alarm 23 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>5 | 1        | For details about the bit definition, see alarm definitions. |
| 64  | [Teleindication]<br>Alarm 24 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>6 | 1        | For details about the bit definition, see alarm definitions. |
| 65  | [Teleindication]<br>Alarm 25 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>7 | 1        | For details about the bit definition, see alarm definitions. |
| 66  | [Teleindication]<br>Alarm 26 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>8 | 1        | For details about the bit definition, see alarm definitions. |
| 67  | [Teleindication]<br>Alarm 27 | RO         | Bitfield<br>16 | N/A  | 1    | 3213<br>9 | 1        | For details about the bit definition, see alarm definitions. |
| 68  | [Teleindication]<br>Alarm 28 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>0 | 1        | For details about the bit definition, see alarm definitions. |
| 69  | [Teleindication]<br>Alarm 29 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>1 | 1        | For details about the bit definition, see alarm definitions. |
| 70  | [Teleindication]<br>Alarm 30 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>2 | 1        | For details about the bit definition, see alarm definitions. |
| 71  | [Teleindication]<br>Alarm 31 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>3 | 1        | For details about the bit definition, see alarm definitions. |

| No. | Signal                       | Read/Write | Type           | Unit | Gain | Address   | Quantity | Scope                                                        |
|-----|------------------------------|------------|----------------|------|------|-----------|----------|--------------------------------------------------------------|
| 72  | [Teleindication]<br>Alarm 32 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>4 | 1        | For details about the bit definition, see alarm definitions. |
| 73  | [Teleindication]<br>Alarm 33 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>5 | 1        | For details about the bit definition, see alarm definitions. |
| 74  | [Teleindication]<br>Alarm 34 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>6 | 1        | For details about the bit definition, see alarm definitions. |
| 75  | [Teleindication]<br>Alarm 35 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>7 | 1        | For details about the bit definition, see alarm definitions. |
| 76  | [Teleindication]<br>Alarm 36 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>8 | 1        | For details about the bit definition, see alarm definitions. |
| 77  | [Teleindication]<br>Alarm 37 | RO         | Bitfield<br>16 | N/A  | 1    | 3214<br>9 | 1        | For details about the bit definition, see alarm definitions. |
| 78  | [Teleindication]<br>Alarm 38 | RO         | Bitfield<br>16 | N/A  | 1    | 3215<br>0 | 1        | For details about the bit definition, see alarm definitions. |
| 79  | [Teleindication]<br>Alarm 39 | RO         | Bitfield<br>16 | N/A  | 1    | 3215<br>1 | 1        | For details about the bit definition, see alarm definitions. |
| 80  | [Teleindication]<br>Alarm 40 | RO         | Bitfield<br>16 | N/A  | 1    | 3215<br>2 | 1        | For details about the bit definition, see alarm definitions. |
| 81  | [Teleindication]<br>Alarm 41 | RO         | Bitfield<br>16 | N/A  | 1    | 3215<br>3 | 1        | For details about the bit definition, see alarm definitions. |



| No. | Signal                       | Read/Write | Type           | Unit | Gain | Address | Quantity | Scope                                                        |
|-----|------------------------------|------------|----------------|------|------|---------|----------|--------------------------------------------------------------|
| 82  | [Teleindication]<br>Alarm 42 | RO         | Bitfield<br>16 | N/A  | 1    | 32154   | 1        | For details about the bit definition, see alarm definitions. |
| 83  | [Teleindication]<br>Alarm 43 | RO         | Bitfield<br>16 | N/A  | 1    | 32155   | 1        | For details about the bit definition, see alarm definitions. |
| 84  | [Teleindication]<br>Alarm 44 | RO         | Bitfield<br>16 | N/A  | 1    | 32156   | 1        | For details about the bit definition, see alarm definitions. |
| 85  | [Teleindication]<br>Alarm 45 | RO         | Bitfield<br>16 | N/A  | 1    | 32157   | 1        | For details about the bit definition, see alarm definitions. |
| 86  | [Teleindication]<br>Alarm 46 | RO         | Bitfield<br>16 | N/A  | 1    | 32158   | 1        | For details about the bit definition, see alarm definitions. |
| 87  | [Teleindication]<br>Alarm 47 | RO         | Bitfield<br>16 | N/A  | 1    | 32159   | 1        | For details about the bit definition, see alarm definitions. |
| 88  | [Teleindication]<br>Alarm 48 | RO         | Bitfield<br>16 | N/A  | 1    | 32160   | 1        | For details about the bit definition, see alarm definitions. |
| 89  | [Teleindication]<br>Alarm 49 | RO         | Bitfield<br>16 | N/A  | 1    | 32161   | 1        | For details about the bit definition, see alarm definitions. |
| 90  | [Teleindication]<br>Alarm 50 | RO         | Bitfield<br>16 | N/A  | 1    | 32162   | 1        | For details about the bit definition, see alarm definitions. |
| 91  | [Teleindication]<br>Alarm 51 | RO         | Bitfield<br>16 | N/A  | 1    | 32163   | 1        | For details about the bit definition, see alarm definitions. |

| No. | Signal                      | Read/Write | Type        | Unit | Gain | Address | Quantity | Scope                                                        |
|-----|-----------------------------|------------|-------------|------|------|---------|----------|--------------------------------------------------------------|
| 92  | [Teleindication] Alarm 52   | RO         | Bitfield 16 | N/A  | 1    | 32164   | 1        | For details about the bit definition, see alarm definitions. |
| 93  | Maximum active power (Pmax) | RO         | U32         | kW   | 1000 | 32853   | 2        |                                                              |
| 94  | Grid code mask 1            | RO         | Bitfield 16 | N/A  | 1    | 32893   | 1        |                                                              |
| 95  | Grid code mask 2            | RO         | Bitfield 16 | N/A  | 1    | 32894   | 1        |                                                              |
| 96  | Grid code mask 3            | RO         | Bitfield 16 | N/A  | 1    | 32895   | 1        |                                                              |
| 97  | Grid code mask 4            | RO         | Bitfield 16 | N/A  | 1    | 32896   | 1        |                                                              |
| 98  | Grid code mask 5            | RO         | Bitfield 16 | N/A  | 1    | 32897   | 1        |                                                              |
| 99  | Grid code mask 6            | RO         | Bitfield 16 | N/A  | 1    | 32898   | 1        |                                                              |
| 100 | Grid code mask 7            | RO         | Bitfield 16 | N/A  | 1    | 32899   | 1        |                                                              |
| 101 | Grid code mask 8            | RO         | Bitfield 16 | N/A  | 1    | 32900   | 1        |                                                              |
| 102 | Grid code mask 9            | RO         | Bitfield 16 | N/A  | 1    | 32901   | 1        |                                                              |
| 103 | Grid code mask 10           | RO         | Bitfield 16 | N/A  | 1    | 32902   | 1        |                                                              |

| No. | Signal                  | Read/Write | Type        | Unit | Gain | Address | Quantity | Scope |
|-----|-------------------------|------------|-------------|------|------|---------|----------|-------|
| 104 | Grid code mask 11       | RO         | Bitfield 16 | N/A  | 1    | 32903   | 1        |       |
| 105 | Grid code mask 12       | RO         | Bitfield 16 | N/A  | 1    | 32904   | 1        |       |
| 106 | Grid code mask 13       | RO         | Bitfield 16 | N/A  | 1    | 32905   | 1        |       |
| 107 | Grid code mask 14       | RO         | Bitfield 16 | N/A  | 1    | 32906   | 1        |       |
| 108 | Grid code mask 15       | RO         | Bitfield 16 | N/A  | 1    | 32907   | 1        |       |
| 109 | Grid code mask 16       | RO         | Bitfield 16 | N/A  | 1    | 32908   | 1        |       |
| 110 | Grid code mask 17       | RO         | Bitfield 16 | N/A  | 1    | 32909   | 1        |       |
| 111 | Power grid code mask 18 | RO         | Bitfield 16 | N/A  | 1    | 32910   | 1        |       |
| 112 | Grid code mask 19       | RO         | Bitfield 16 | N/A  | 1    | 32911   | 1        |       |
| 113 | Grid code mask 20       | RO         | Bitfield 16 | N/A  | 1    | 32912   | 1        |       |
| 114 | Grid code mask 21       | RO         | Bitfield 16 | N/A  | 1    | 32913   | 1        |       |
| 115 | Grid code mask 22       | RO         | Bitfield 16 | N/A  | 1    | 32914   | 1        |       |
| 116 | Grid code mask 23       | RO         | Bitfield 16 | N/A  | 1    | 32915   | 1        |       |

| No. | Signal                        | Read/Write | Type        | Unit | Gain | Address | Quantity | Scope |
|-----|-------------------------------|------------|-------------|------|------|---------|----------|-------|
| 117 | Power grid code mask 24       | RO         | Bitfield 16 | N/A  | 1    | 32916   | 1        |       |
| 118 | Grid code mask 25             | RO         | Bitfield 16 | N/A  | 1    | 32917   | 1        |       |
| 119 | Grid code mask 26             | RO         | Bitfield 16 | N/A  | 1    | 32918   | 1        |       |
| 120 | Grid code mask 27             | RO         | Bitfield 16 | N/A  | 1    | 32919   | 1        |       |
| 121 | Power grid code mask 28       | RO         | Bitfield 16 | N/A  | 1    | 32920   | 1        |       |
| 122 | Grid code mask 29             | RO         | Bitfield 16 | N/A  | 1    | 32921   | 1        |       |
| 123 | Grid code mask 30             | RO         | Bitfield 16 | N/A  | 1    | 32922   | 1        |       |
| 124 | Grid code mask 31             | RO         | Bitfield 16 | N/A  | 1    | 32923   | 1        |       |
| 125 | Grid code mask 32             | RO         | Bitfield 16 | N/A  | 1    | 32924   | 1        |       |
| 126 | AB line voltage of power grid | RO         | U16         | V    | 10   | 32970   | 1        |       |
| 127 | BC line voltage of power grid | RO         | U16         | V    | 10   | 32971   | 1        |       |
| 128 | CA line voltage of power grid | RO         | U16         | V    | 10   | 32972   | 1        |       |
| 129 | Phase A voltage of power grid | RO         | U16         | V    | 10   | 32973   | 1        |       |
| 130 | Phase B voltage of power grid | RO         | U16         | V    | 10   | 32974   | 1        |       |

| No. | Signal                                                  | Read/Write | Type | Unit  | Gain | Address | Quantity | Scope |
|-----|---------------------------------------------------------|------------|------|-------|------|---------|----------|-------|
| 131 | Phase C voltage of power grid                           | RO         | U16  | V     | 10   | 32975   | 1        |       |
| 132 | Phase A current of power grid                           | RO         | I32  | A     | 1000 | 32976   | 2        |       |
| 133 | Phase B current of power grid                           | RO         | I32  | A     | 1000 | 32978   | 2        |       |
| 134 | Phase C current of power grid                           | RO         | I32  | A     | 1000 | 32980   | 2        |       |
| 135 | Active power                                            | RO         | I32  | kW    | 1000 | 32986   | 2        |       |
| 136 | Reactive power                                          | RO         | I32  | kV ar | 1000 | 33000   | 2        |       |
| 137 | Power factor                                            | RO         | I16  | N/A   | 1000 | 33020   | 1        |       |
| 138 | Grid Frequency                                          | RO         | U16  | Hz    | 100  | 33021   | 1        |       |
| 139 | DC voltage                                              | RO         | U16  | V     | 10   | 33024   | 1        |       |
| 140 | DC current                                              | RO         | I32  | A     | 100  | 33025   | 2        |       |
| 141 | Maximum active power of reverse rectification ( RPmax ) | RO         | I32  | kW    | 1000 | 33097   | 2        |       |
| 142 | Energy discharged today                                 | RO         | U32  | kW h  | 100  | 33152   | 2        |       |
| 143 | Energy charged today                                    | RO         | U32  | kW h  | 100  | 33154   | 2        |       |
| 144 | Energy discharged this month                            | RO         | U32  | kW h  | 100  | 33156   | 2        |       |
| 145 | Energy charged this month                               | RO         | U32  | kW h  | 100  | 33158   | 2        |       |

| No. | Signal                            | Read/Write | Type | Unit | Gain | Address                          | Quantity | Scope                                       |
|-----|-----------------------------------|------------|------|------|------|----------------------------------|----------|---------------------------------------------|
| 146 | Energy discharged this year       | RO         | U32  | kWh  | 100  | 33160                            | 2        |                                             |
| 147 | Energy charged this year          | RO         | U32  | kWh  | 100  | 33162                            | 2        |                                             |
| 148 | Total energy discharged           | RO         | U32  | kWh  | 100  | 33164                            | 2        |                                             |
| 149 | Total energy charged              | RO         | U32  | kWh  | 100  | 33166                            | 2        |                                             |
| 150 | Actual charging power capability  | RO         | U32  | kW   | 1000 | 33215                            | 2        | PCS Actual charging power capability value  |
| 151 | Actual discharge power capability | RO         | U32  | kW   | 1000 | 33217                            | 2        | PCS Actual discharge power capability value |
| 152 | Voltage of Pack ( Pack1-4 )       | RO         | U16  | V    | 10   | 33662<br>33862<br>34062<br>34262 | 1        |                                             |
| 153 | SOC ( Pack1-4 )                   | RO         | U16  | %    | 1    | 33663<br>33863<br>34063<br>34263 | 1        | [0,100]                                     |

| No. | Signal                                  | Read/Write | Type | Unit | Gain | Address                      | Quantity | Scope                                                                      |
|-----|-----------------------------------------|------------|------|------|------|------------------------------|----------|----------------------------------------------------------------------------|
| 154 | SOH ( Pack1-4 )                         | RO         | U16  | %    | 1    | 3364<br>3386<br>3406<br>3426 | 1        | [0,100]                                                                    |
| 155 | System Time                             | RW         | U32  | N/A  | 1    | 40000                        |          | Epoch second UTC                                                           |
| 156 | City                                    | RW         | U32  | N/A  | 1    | 40002                        |          |                                                                            |
| 157 | Daylight saving time enable             | RW         | U16  | N/A  | 1    | 40004                        |          | 0: Disable<br>1: Enable                                                    |
| 158 | Power-on or power-off of Energy storage | RW         | U16  | N/A  | 1    | 42000                        | 1        | 1: Run<br>2: Off                                                           |
| 159 | Charging cut-off SOC                    | RW         | U16  | %    | 1    | 42002                        | 1        | [90,100]                                                                   |
| 160 | Discharge cut-off SOC                   | RW         | U16  | %    | 1    | 42003                        | 1        | [0,15]                                                                     |
| 161 | Charging confirmation                   | WO         | U16  | N/A  | 1    | 42207                        | 1        | 0: Not allowed,<br>1: Allows                                               |
| 162 | Grid code                               | RW         | U16  | N/A  | 1    | 42921                        | 1        | For detailed definition of scope, please refer to the "Grid Standard Code" |
| 163 | Working mode                            | RW         | U16  | N/A  | 1    | 43133                        | 1        | 0: PQ<br>1: VSG                                                            |
| 164 | Active power (%) 【 high 】               | RW         | I16  | %    | 100  | 42913                        | 1        | [-100.00,100.00]                                                           |
| 165 | Reactive power compensation (Q/S)       | RW         | I16  | %    | 100  | 42914                        | 1        | [-100.00,100.00]                                                           |
| 166 | Active power (kW)                       | RW         | I32  | kW   | 1000 | 42915                        | 2        | [Rpmax, Pmax]                                                              |

| No. | Signal                | Re<br>ad/<br>Wr<br>ite | Typ<br>e | Un<br>it | Gai<br>n | Addr<br>ess | Qua<br>ntit<br>y | Scope                                             |
|-----|-----------------------|------------------------|----------|----------|----------|-------------|------------------|---------------------------------------------------|
| 167 | Power factor          | RW                     | I16      | N/A      | 1000     | 42917       | 1                | (-1.000,-0.800]∪<br>[0.800,1.000]                 |
| 168 | Active power baseline | RW                     | U32      | kW       | 1000     | 42936       | 2                | As a benchmark for active scheduling (percentage) |

### 3.2 Alarm Definitions

Table 3-2 Alarm definition table

| ID   | Alarm Name                                           | Register Address | Bit |
|------|------------------------------------------------------|------------------|-----|
| 3100 | Battery Pack and External Power Connected Abnormally | 32133            | 0   |
| 3101 | Battery Pack Lifespan Reached                        | 32133            | 1   |
| 3102 | Battery Pack Overvoltage                             | 32133            | 2   |
| 3103 | Battery Pack Undervoltage                            | 32133            | 3   |
| 3104 | Battery Overcurrent Protection                       | 32133            | 4   |
| 3105 | Battery Pack Overtemperature                         | 32133            | 5   |
| 3106 | Battery Pack Undertemperature                        | 32133            | 6   |
| 3107 | Battery Pack Locked                                  | 32133            | 7   |
| 3108 | Battery Faulty                                       | 32133            | 8   |
| 3109 | Battery Pack SOH Low                                 | 32133            | 9   |
| 3112 | Power Connection Sampling of Battery Pack Abnormal   | 32133            | 12  |
| 3113 | Battery Pack Configuration Data Abnormal             | 32133            | 13  |
| 3114 | Battery Pack and System Specifications Mismatched    | 32133            | 14  |



| ID        | Alarm Name                                                                 | Register Address | Bit  |
|-----------|----------------------------------------------------------------------------|------------------|------|
| 3115      | Cell Temperature Rises Abnormally                                          | 32133            | 15   |
| 3116      | Pack Thermal Runaway                                                       | 32134            | 0    |
| 3117      | Battery Pack Charging Failed                                               | 32134            | 1    |
| 3118–3131 | Reserved                                                                   | 32134            | 2–15 |
| 3132–3147 | Reserved                                                                   | 32135            | 0–15 |
| 3148–3160 | Reserved                                                                   | 32136            | 0–12 |
| 3161      | Balancing Module Overcurrent Protection                                    | 32136            | 13   |
| 3162      | Balancing Module Internal Error                                            | 32136            | 14   |
| 3163      | Balancing Module Bus Voltage Abnormal                                      | 32136            | 15   |
| 3164      | Balancing Module Battery Voltage Abnormal                                  | 32137            | 0    |
| 3165      | Overtemperature Protection Inside the Balancing Module                     | 32137            | 1    |
| 3166      | MCU Overtemperature Protection on the Battery Side of the Balancing Module | 32137            | 2    |
| 3167      | Overtemperature Protection for the Wiring Terminal of the Balancing Module | 32137            | 3    |
| 3168      | Reserved                                                                   | 32137            | 4    |
| 3169      | Balancing Module Bus Soft-Start Failed                                     | 32137            | 5    |
| 3170      | Balancing Module Version Mismatch                                          | 32137            | 6    |
| 3171      | Balancing Module Address Pairing Timeout                                   | 32137            | 7    |
| 3172      | Soft-Start MOS of Balancing Module Short-Circuited                         | 32137            | 8    |
| 3173      | Balancing Module Abnormal                                                  | 32137            | 9    |

| ID        | Alarm Name                                     | Register Address | Bit   |
|-----------|------------------------------------------------|------------------|-------|
| 3174      | Balancing Module Overcurrent Fault             | 32137            | 10    |
| 3175–3179 | Reserved                                       | 32137            | 11–15 |
| 3180–3195 | Reserved                                       | 32138            | 0–15  |
| 3196–3211 | Reserved                                       | 32139            | 0–15  |
| 3212–3221 | Reserved                                       | 32140            | 0–9   |
| 3222      | BMU Internal Short Circuit                     | 32140            | 10    |
| 3223      | BMU Sampling Cable Disconnected                | 32140            | 11    |
| 3224      | BMU Faulty                                     | 32140            | 12    |
| 3225      | Reserved                                       | 32140            | 13    |
| 3226      | Disconnection Between BMUs                     | 32140            | 14    |
| 3227      | Reserved                                       | 32140            | 15    |
| 3228      | Reserved                                       | 32141            | 0     |
| 3229      | BMU Communication Failure                      | 32141            | 1     |
| 3230–3243 | Reserved                                       | 32141            | 2–15  |
| 3244–3259 | Reserved                                       | 32142            | 0–15  |
| 3260–3275 | Reserved                                       | 32143            | 0–15  |
| 3276–3291 | Reserved                                       | 32144            | 0–15  |
| 3292–3299 | Reserved                                       | 32145            | 0–7   |
| 3300      | RPCB Voltage Abnormal                          | 32145            | 8     |
| 3301      | RPCB Port Short-Circuited                      | 32145            | 9     |
| 3302      | Internal RPCB Temperature Abnormal             | 32145            | 10    |
| 3303      | RPCB Overcurrent Fault                         | 32145            | 11    |
| 3304      | Battery-Side ISO Insulation Detection Abnormal | 32145            | 12    |
| 3305      | RPCB Power Loop Abnormal                       | 32145            | 13    |
| 3306      | RPCB Version Mismatch                          | 32145            | 14    |
| 3307      | RPCB Abnormal                                  | 32145            | 15    |

| ID        | Alarm Name                                                        | Register Address | Bit  |
|-----------|-------------------------------------------------------------------|------------------|------|
| 3308      | Battery Rack Residual Current Abnormal                            | 32146            | 0    |
| 3309      | RPCB Overcurrent Protection Triggered                             | 32146            | 1    |
| 3310      | RPCB Fan Alarm                                                    | 32146            | 2    |
| 3311      | SPD Faulty                                                        | 32146            | 3    |
| 3312      | Battery-Side ISO Insulation Detection Alarm                       | 32146            | 4    |
| 3313      | Auxiliary Power Abnormal                                          | 32146            | 5    |
| 3314      | RPCB Overcurrent Alarm                                            | 32146            | 6    |
| 3315~3323 | Reserved                                                          | 32146            | 7~15 |
| 3324~3339 | Reserved                                                          | 32147            | 0~15 |
| 3340~3350 | Reserved                                                          | 32148            | 0~10 |
| 3352      | Abnormal Total Voltage of Battery Rack                            | 32148            | 12   |
| 3353      | Total Voltage of Battery Rack and Total Cell Voltage Inconsistent | 32148            | 13   |
| 3354      | Battery Pack Mixed Use Failed                                     | 32148            | 14   |
| 3355      | BCU Chip Overtemperature                                          | 32148            | 15   |
| 3356      | BCU Internal Exception                                            | 32149            | 0    |
| 3357      | Balancing Module Software Versions Inconsistent                   | 32149            | 1    |
| 3358      | BCU Auxiliary Power Abnormal                                      | 32149            | 2    |
| 3359      | BCU Memory Abnormal                                               | 32149            | 3    |
| 3360      | RPCB Communication Failure                                        | 32149            | 4    |
| 3361      | DCDC Communication Failure                                        | 32149            | 5    |
| 3362      | PCS Communication Failure                                         | 32149            | 6    |
| 3363      | Balancing Module Communication Failure                            | 32149            | 7    |
| 3364~3368 | Reserved                                                          | 32149            | 8~12 |

| ID        | Alarm Name                                           | Register Address | Bit  |
|-----------|------------------------------------------------------|------------------|------|
| 3369      | ESS SOH Calibration                                  | 32149            | 13   |
| 3370      | Reserved                                             | 32149            | 14   |
| 3371      | Battery Rack Voltage Exceeding Threshold             | 32149            | 15   |
| 3372      | Reserved                                             | 32150            | 0    |
| 3373      | Battery Pack Sampling Abnormal                       | 32150            | 1    |
| 3374      | Reserved                                             | 32150            | 2    |
| 3375      | Battery Voltage Inconsistent                         | 32150            | 3    |
| 3376      | Battery Temperature Inconsistent                     | 32150            | 4    |
| 3377~3387 | Reserved                                             | 32150            | 5~15 |
| 3388~3399 | Reserved                                             | 32151            | 0~11 |
| 3400      | DCDC Protection                                      | 32151            | 12   |
| 3401      | DCDC Faulty                                          | 32151            | 13   |
| 3402      | Overvoltage Protection for DCDC Bus Port             | 32151            | 14   |
| 3403      | DCDC Overhumidity Protection                         | 32151            | 15   |
| 3404      | DCDC Overtemperature Protection                      | 32152            | 0    |
| 3405      | DCDC Undertemperature Protection                     | 32152            | 1    |
| 3406      | Overvoltage Protection for DCDC Bus Terminal         | 32152            | 2    |
| 3407      | Overtemperature Protection for DCDC Battery Terminal | 32152            | 3    |
| 3408      | DCDC Liquid Cooling Abnormal                         | 32152            | 4    |
| 3409      | DCDC Versions Mismatched                             | 32152            | 5    |
| 3410      | DCDC NTC Faulty                                      | 32152            | 6    |
| 3411      | DCDC DSP_EPO Faulty                                  | 32152            | 7    |
| 3412      | DCDC Short Circuit Protection                        | 32152            | 8    |

| ID        | Alarm Name                                | Register Address | Bit   |
|-----------|-------------------------------------------|------------------|-------|
| 3413      | DCDC Bus Overvoltage                      | 32152            | 9     |
| 3414~3419 | Reserved                                  | 32152            | 10~15 |
| 3420~3435 | Reserved                                  | 32153            | 0~15  |
| 3436~3451 | Reserved                                  | 32154            | 0~15  |
| 3452~3467 | Reserved                                  | 32155            | 0~15  |
| 3468~3483 | Reserved                                  | 32156            | 0~15  |
| 3484~3499 | Reserved                                  | 32157            | 0~15  |
| 3500      | PCS DC Overvoltage                        | 32158            | 0     |
| 3501      | PCS DC Bus in Reverse Polarity            | 32158            | 1     |
| 3502      | Reserved                                  | 32158            | 2     |
| 3503      | PCS Grid Phase Wire Short-Circuited to PE | 32158            | 3     |
| 3504      | PCS Grid Failed                           | 32158            | 4     |
| 3505      | PCS Grid Undervoltage                     | 32158            | 5     |
| 3506      | PCS Grid Overvoltage                      | 32158            | 6     |
| 3507      | PCS Grid Voltage Imbalanced               | 32158            | 7     |
| 3508      | PCS Grid Overfrequency                    | 32158            | 8     |
| 3509      | PCS Grid Underfrequency                   | 32158            | 9     |
| 3510      | PCS Grid Frequency Unstable               | 32158            | 10    |
| 3511      | PCS AC Overcurrent                        | 32158            | 11    |
| 3512      | PCS DC Component Overhigh                 | 32158            | 12    |
| 3513      | Reverse Phase Sequence on PCS AC Side     | 32158            | 13    |
| 3514      | PCS Residual Current Abnormal             | 32158            | 14    |
| 3515      | PCS Grounding Abnormal                    | 32158            | 15    |
| 3516      | Low PCS Insulation Resistance             | 32159            | 0     |
| 3517      | PCS Temperature High                      | 32159            | 1     |

| ID        | Alarm Name                                        | Register Address | Bit   |
|-----------|---------------------------------------------------|------------------|-------|
| 3518      | PCS Abnormal                                      | 32159            | 2     |
| 3519      | PCS Update Failed or Versions Mismatched          | 32159            | 3     |
| 3520      | PCS Internal Fan Abnormal                         | 32159            | 4     |
| 3521      | PCS AC Terminal Temperature Abnormal              | 32159            | 5     |
| 3522      | PCS DC Terminal Temperature Abnormal              | 32159            | 6     |
| 3523      | PCS Black Start Failed                            | 32159            | 7     |
| 3524      | Incorrect Black Start Instruction Sequence of PCS | 32159            | 8     |
| 3525      | PCS Fuse Broken                                   | 32159            | 9     |
| 3526      | PCS Fuse Self-Check Abnormal                      | 32159            | 10    |
| 3527      | PCS FAST I/O Self-Test Abnormal                   | 32159            | 11    |
| 3528      | PCS DC Bus Short-Circuited                        | 32159            | 12    |
| 3529      | PCS Relay Overtemperature                         | 32159            | 13    |
| 3530      | PCS AC Resonance                                  | 32159            | 14    |
| 3531      | PCS Derated                                       | 32159            | 15    |
| 3530~3531 | Reserved                                          | 32159            | 14~15 |
| 3532~3547 | Reserved                                          | 32160            | 0~15  |
| 3548~3563 | Reserved                                          | 32161            | 0~15  |
| 3564~3579 | Reserved                                          | 32162            | 0~15  |
| 3580~3595 | Reserved                                          | 32163            | 0~15  |
| 3596~3599 | Reserved                                          | 32164            | 0~3   |
| 3600      | Power loss alarm                                  | 30455            | 0     |
| 3601      | Power voltage abnormal                            | 30455            | 1     |
| 3602      | Power frequency abnormal                          | 30455            | 2     |
| 3603      | Outdoor temperature sensor fault                  | 30455            | 3     |
| 3604      | Outdoor low temperature alarm                     | 30455            | 4     |

| ID        | Alarm Name                                  | Register Address | Bit   |
|-----------|---------------------------------------------|------------------|-------|
| 3605      | LTMS communication abnormal                 | 30455            | 5     |
| 3606      | LTMS expiration alarm                       | 30455            | 6     |
| 3607      | Reserved                                    | 30455            | 7     |
| 3608      | Certificate about to expire                 | 30455            | 8     |
| 3609      | Certificate has expired                     | 30455            | 9     |
| 3610~3615 | Reserved                                    | 30455            | 10~15 |
| 3616~3619 | Reserved                                    | 30456            | 0~3   |
| 3620      | Compressor discharge high pressure alarm    | 30456            | 4     |
| 3621      | Compressor suction low pressure alarm       | 30456            | 5     |
| 3622      | Compressor low superheat degree alarm       | 30456            | 6     |
| 3623      | Compressor discharge pressure sensor fault  | 30456            | 7     |
| 3624      | Condenser outlet pressure sensor fault      | 30456            | 8     |
| 3625      | Condenser outlet temperature sensor fault   | 30456            | 9     |
| 3626      | Compressor suction pressure sensor fault    | 30456            | 10    |
| 3627      | Compressor suction temperature sensor fault | 30456            | 11    |
| 3628      | Dehumidifying temperature sensor fault      | 30456            | 12    |
| 3629~3631 | Reserved                                    | 30456            | 13~15 |
| 3632~3639 | Reserved                                    | 30457            | 0~7   |
| 3640      | Compressor drive alarm                      | 30457            | 8     |
| 3641      | Compressor drive output abnormal            | 30457            | 9     |
| 3642      | Compressor overcurrent alarm                | 30457            | 10    |
| 3643      | Compressor drive communication abnormal     | 30457            | 11    |

| ID        | Alarm Name                                                 | Register Address | Bit   |
|-----------|------------------------------------------------------------|------------------|-------|
| 3644      | High discharge temperature alarm                           | 30457            | 12    |
| 3645      | Compressor Discharge Temperature Sensor Faulty             | 30457            | 13    |
| 3646      | Insufficient Refrigerant                                   | 30457            | 14    |
| 3647      | Reserved                                                   | 30457            | 15    |
| 3648~3649 | Reserved                                                   | 30458            | 0~1   |
| 3650      | Insufficient Cooling Capacity                              | 30458            | 2     |
| 3651~3654 | Reserved                                                   | 30458            | 3~6   |
| 3655      | Auxiliary power abnormal                                   | 30458            | 7     |
| 3656~3659 | Reserved                                                   | 30458            | 8~11  |
| 3660      | Outdoor cooling module blocked                             | 30458            | 12    |
| 3661      | Outdoor heat exchanger temperature sensor fault            | 30458            | 13    |
| 3662~3663 | Reserved                                                   | 30458            | 14~15 |
| 3664      | Reserved                                                   | 30459            | 0     |
| 3665      | Outdoor fan fault                                          | 30459            | 1     |
| 3666      | Dehumidifier fan fault                                     | 30459            | 2     |
| 3667~3674 | Reserved                                                   | 30459            | 3~10  |
| 3675      | Electric heater fault                                      | 30459            | 11    |
| 3676      | Electric heater power overvoltage alarm                    | 30459            | 12    |
| 3677~3679 | Reserved                                                   | 30459            | 13~15 |
| 3680      | Power-side supply water temperature sensor fault           | 30460            | 0     |
| 3681      | Power-side return water temperature sensor fault           | 30460            | 1     |
| 3682      | Power-side supply/return water temperature sensor abnormal | 30460            | 2     |
| 3683      | Battery-side supply water temperature sensor fault         | 30460            | 3     |



| ID        | Alarm Name                                                   | Register Address | Bit   |
|-----------|--------------------------------------------------------------|------------------|-------|
| 3684      | Battery-side return water temperature sensor fault           | 30460            | 4     |
| 3685      | Battery-side supply/return water temperature sensor abnormal | 30460            | 5     |
| 3686      | Battery-side supply water high temperature alarm             | 30460            | 6     |
| 3687      | Battery-side supply water low temperature alarm              | 30460            | 7     |
| 3688      | Coolant expiration alarm                                     | 30460            | 8     |
| 3689      | Shutdown due to coolant expiration                           | 30460            | 9     |
| 3690      | Coolant replacement not completed                            | 30460            | 10    |
| 3691~3695 | Reserved                                                     | 30460            | 11~15 |
| 3696~3704 | Reserved                                                     | 30461            | 0~8   |
| 3705      | Water pump power supply abnormal                             | 30461            | 9     |
| 3706      | Water pump function abnormal                                 | 30461            | 10    |
| 3707      | Water pump fault                                             | 30461            | 11    |
| 3708~3711 | Reserved                                                     | 30461            | 12~15 |
| 3712~3714 | Reserved                                                     | 30462            | 0~2   |
| 3715      | Multi-way valve communication abnormal                       | 30462            | 3     |
| 3716      | Multi-way valve power supply abnormal                        | 30462            | 4     |
| 3717      | Multi-way valve faulty                                       | 30462            | 5     |
| 3718~3724 | Reserved                                                     | 30462            | 6~12  |
| 3725      | Water tank low liquid level alarm                            | 30462            | 13    |
| 3726      | Reserved                                                     | 30462            | 14    |
| 3727      | Reserved                                                     | 30462            | 15    |
| 3728~3743 | Reserved                                                     | 30463            | 0~15  |
| 3744~3759 | Reserved                                                     | 30464            | 0~15  |

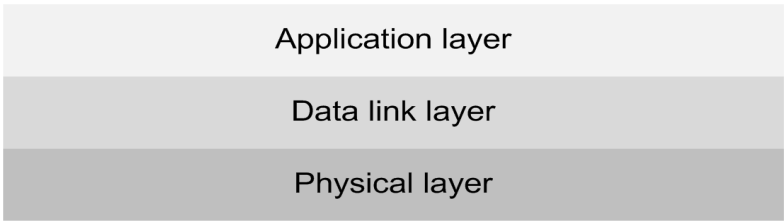
| ID        | Alarm Name                                              | Register Address | Bit  |
|-----------|---------------------------------------------------------|------------------|------|
| 3760~3775 | Reserved                                                | 30465            | 0~15 |
| 3776~3791 | Reserved                                                | 30466            | 0~15 |
| 3792~3799 | Reserved                                                | 30510            | 0~7  |
| 3880      | AC SPD Faulty                                           | 30510            | 8    |
| 3881      | Door Status Alarm                                       | 30510            | 9    |
| 3882      | ESS Door Open                                           | 30510            | 10   |
| 3883      | Water Alarm                                             | 30510            | 11   |
| 3884      | Smoke Detector Alarm                                    | 30510            | 12   |
| 3885      | High Concentration of Combustible Gas                   | 30510            | 13   |
| 3886      | Combustible Gas Detector Communication Failed           | 30510            | 14   |
| 3887      | Combustible Gas Detector Faulty                         | 30510            | 15   |
| 3888      | Temperature and Humidity Sensor Communication Failed    | 30511            | 0    |
| 3889      | Temperature and Humidity Sensor Faulty                  | 30511            | 1    |
| 3890      | Heat Detector Alarm                                     | 30511            | 2    |
| 3891      | High Ambient Temperature Inside ESS Cabin               | 30511            | 3    |
| 3892      | EPO Alarm                                               | 30511            | 4    |
| 3893      | Fire Alarm                                              | 30511            | 5    |
| 3894      | Exhaust Fan Faulty                                      | 30511            | 6    |
| 3895      | Devices Connected and System Configuration Inconsistent | 30511            | 7    |
| 3896~3897 | Reserved                                                | 30511            | 8~9  |
| 3898      | TRSD Abnormal                                           | 30511            | 10   |
| 3899      | TRSD Valve Open                                         | 30511            | 11   |
| 3900      | High Relative Humidity Inside ESS Cabin                 | 30511            | 12   |

| ID        | Alarm Name                                         | Register Address | Bit   |
|-----------|----------------------------------------------------|------------------|-------|
| 3901      | Offering Software Update Package Not Backed Up     | 30511            | 13    |
| 3902      | Component Software Versions Inconsistent           | 30511            | 14    |
| 3903      | E-label Board Data Abnormal                        | 30511            | 15    |
| 3904      | Certificate About to Expire                        | 30512            | 0     |
| 3905      | Certificate Expired                                | 30512            | 1     |
| 3906      | Communication with Upper-layer Controller Abnormal | 30512            | 2     |
| 3907      | Reserved                                           | 30512            | 3     |
| 3908      | Sensor End of Life                                 | 30512            | 4     |
| 3909      | TRSD Communication Abnormal                        | 30512            | 5     |
| 3910      | Auxiliary Power Meter Communication Abnormal       | 30512            | 6     |
| 3911      | Display Module Communication Failure               | 30512            | 7     |
| 3912      | Startup Authorization Not Obtained                 | 30512            | 8     |
| 3913      | Fire Extinguishing Agents in TRSD Sprayed          | 30512            | 9     |
| 3914~3919 | Reserved                                           | 30512            | 10~15 |
| 3920~3935 | Reserved                                           | 30513            | 0~15  |
| 3936~3951 | Reserved                                           | 30514            | 0~15  |
| 3952~3967 | Reserved                                           | 30515            | 0~15  |
| 3968~3983 | Reserved                                           | 30516            | 0~15  |
| 3984~3999 | Reserved                                           | 30517            | 0~15  |

# 4 Overview of the Communications Protocol

The Modbus communications protocol consists of the following layers.

Figure 4-1 Modbus protocol layers



- 4.1 Physical Layer
- 4.2 Data Link Layer
- 4.3 Application Layer

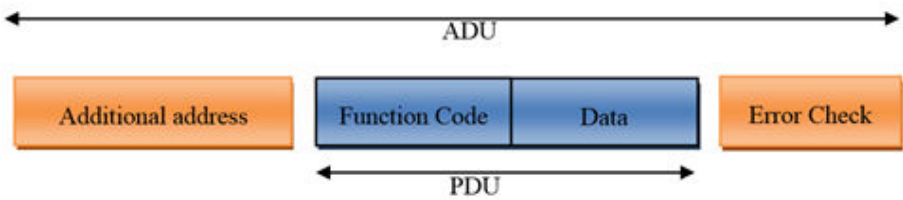
## 4.1 Physical Layer

Huawei ESS containers provide Modbus communication based on FE physical media. The communication is based on the TCP link and complies with the Modbus-TCP format.

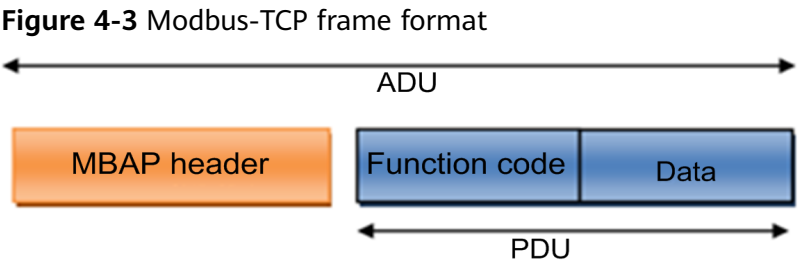
## 4.2 Data Link Layer

The following figure shows the generic frame structure of the Modbus protocol.

Figure 4-2 Modbus generic frame format



## 4.2.1 Modbus-TCP



### 4.2.1.1 ADU Length

The recommended frame length is 260 bytes based on the standard. When some extended functions are applied, the data service provider may extend the ADU to a proper length based on available resources to improve network transmission efficiency. The ADU length is indicated by the length field in the MBAP header.

### 4.2.1.2 MBAP Header

When Modbus runs on top of TCP/IP, a dedicated MBAP header (Modbus application protocol header) is used to identify the Modbus ADU. The Modbus header consists of four fields and seven bytes, which are defined as follows.

Table 4-1 MBAP definition

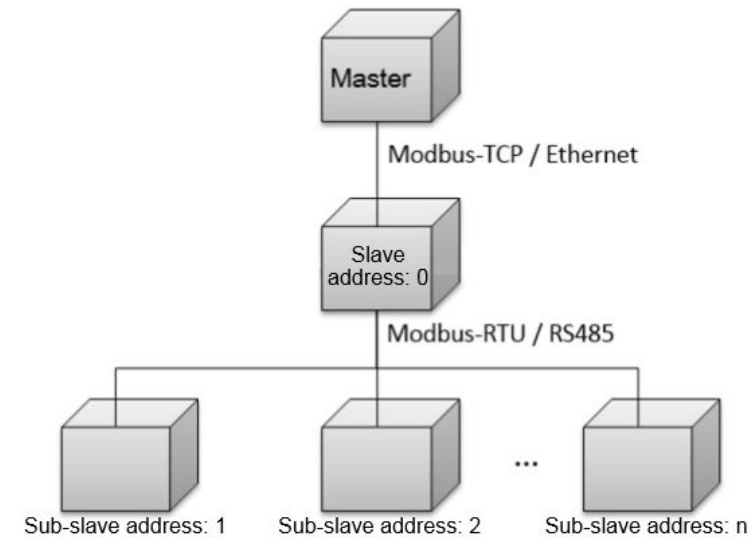
| Data Field              | Length (Byte) | Description                                                      | Client                                                                                     | Server                                                                                                            |
|-------------------------|---------------|------------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Transmission identifier | 2             | Matching identifier between a request frame and a response frame | Assigned by the client. It is recommended that each frame be assigned a unique identifier. | The identifier of the response frame from the server must be the same as that of the corresponding request frame. |
| Protocol                | 2             | 0 = Modbus protocol                                              | Assigned by the client; 0 by default                                                       | The identifier of the response frame from the server must be the same as that of the corresponding request frame. |

| Data Field        | Length (Byte) | Description           | Client                                                        | Server                                                                                                            |
|-------------------|---------------|-----------------------|---------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|
| Data length       | 2             | Follow-up data length | Assigned by the client based on the actual data frame         | Assigned by the server based on the actual frame length                                                           |
| Logical device ID | 1             | 0                     | Assigned by the client based on the actual data frame request | The identifier of the response frame from the server must be the same as that of the corresponding request frame. |

4.2.1.3 Communications Address

Based on the TCP communications host, unit 0 is used by default to access the directly connected slave node, and other addresses are used to access the downstream devices of the slave node. The default address of the slave node is 0. The address is configurable.

Figure 4-4 Three-layer communications addresses



4.2.1.4 TCP Port

In a local area network or VPN environment, the master node may initiate a TCP socket connection to the slave node. The master node can use port 502 to request data services from the slave node.

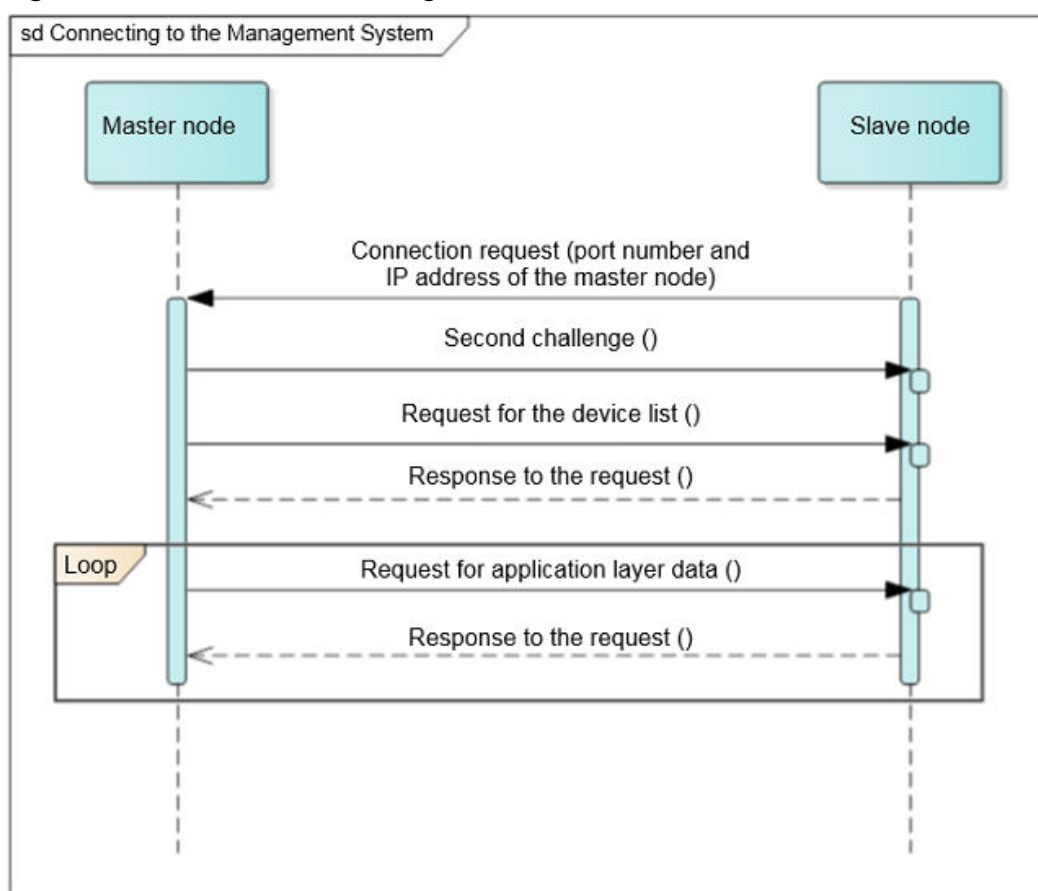
In a non-VPN environment across the public network, the device deployed on the internal network needs to initiate TCP socket link establishment to the master node exposed on the public network. In this case, you need to specify the port number of the master node on the slave node. To ensure security and minimize traffic, the master node must provide at least one encrypted port and one non-encrypted port.

#### 4.2.1.5 TCP Link Establishment Process

This section focuses on the cross-public network application.

The following figure shows the process of connecting a slave node.

**Figure 4-5** Process of establishing a secure TCP connection



### 4.3 Application Layer

### 4.3.1 Function Code

Table 4-2 Function code

| Function Code | Description                | Remarks                                                                 |
|---------------|----------------------------|-------------------------------------------------------------------------|
| 0x03          | Reads registers.           | Supports continuous reading of a single register or multiple registers. |
| 0x06          | Writes a single register.  | Supports writing into a single register.                                |
| 0x10          | Writes multiple registers. | Supports continuous writing into multiple registers.                    |

### 4.3.2 Exception Code

Exception codes must be unique for each network element (NE) type. The names and descriptions should be provided in the NE interface document. Different versions of the same NE type must be backward compatible. Exception codes in use cannot be assigned to other exceptions.

Table 4-3 Exception codes returned by an NE (0x00–0x8F used for common exception codes)

| Code | Term             | Description                                                                                                                                                                                                                                                                                                                                                                                                                     |
|------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x01 | Invalid function | The function code received in the query is not an allowable action for the server (or slave node). This may be because the function code is only applicable to newer devices, and cannot be implemented in the unit selected. It also indicates that the server (or slave node) is in the wrong state to process a request of this type, for example because it is not configured and is being asked to return register values. |



| Code | Term                                      | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|------|-------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 0x02 | Invalid data address                      | The data address received in the query is not an allowable address for the server. More specifically, the combination of the starting register address and register quantity is invalid. For a controller with 100 registers, the PDU addresses the first register as 0 and the last one as 99. If the starting register address in a request is 96 and the register quantity is 4, the request can obtain the return values of registers 96, 97, 98, and 99. If the starting register address of a request is 96 and the register quantity is 5, the request fails and the exception code 0x02 "Invalid data address" is returned because the request attempts to read registers 96, 97, 98, 99, and 100, among which 100 is not a defined address. |
| 0x03 | Invalid data value                        | The value contained in the query data field is not an allowable value for the server (or slave node). The value indicates a fault in the structure of the remainder of a complex request, such as an incorrectly implied length. It does not mean that a data item submitted for storage in a register has a value outside the expectation of the application program since the Modbus protocol is unaware of the significance of any particular value of any particular register.                                                                                                                                                                                                                                                                   |
| 0x04 | Slave node failure                        | An error occurs while the server attempts to perform the requested action.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 0x06 | Slave node busy                           | The server cannot accept a Modbus request PDU. The client application determines whether and when to retransmit the request.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| 0x80 | No permission                             | An operation is not allowed because of a permission authentication failure or permission expiration.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 0x90 | Southbound access device response timeout | The response from the southbound device times out or the communication is disconnected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 0x91 | Internal unit response timeout            | The response from the internal unit times out or the communication is disconnected.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |

### 4.3.3 Read Registers (0x03)

### 4.3.3.1 Frame Format of a Master Node Request

| Data Field                | Length  | Description   |
|---------------------------|---------|---------------|
| Function Code             | 1 byte  | 0x03          |
| Starting Register Address | 2 bytes | 0x0000–0xFFFF |
| Number of Registers       | 2 bytes | 1–125         |

### 4.3.3.2 Frame Format of a Normal Response from a Slave Node

| Data Field     | Length             | Description  |
|----------------|--------------------|--------------|
| Function Code  | 1 byte             | 0x03         |
| Byte Count     | 1 byte             | 2 x <i>N</i> |
| Register Value | 2 x <i>N</i> bytes | N/A          |

 **NOTE**

*N* refers to the number of registers.

### 4.3.3.3 Frame Format of an Abnormal Response from a Slave Node

| Data Field     | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Function Code  | 1 byte | 0x83                                       |
| Exception Code | 1 byte | See <a href="#">4.3.2 Exception Code</a> . |

### 4.3.3.4 Examples

This section takes the Modbus-TCP communications frames as an example. The differences between Modbus-RTU and Modbus-TCP lie in the additional address field and the CRC. Pay attention to the differences when using the Modbus-RTU frames. This also works for the follow-up examples.

The master node sends a query request (register address: 32306/0X7E32) to the slave node (logical device ID: 00).

| Description |                     | Frame Data |
|-------------|---------------------|------------|
| MBAP Header | Protocol identifier | 00         |
|             |                     | 01         |
|             | Protocol            | 00         |

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 06         |
|               | Logical device ID   | 00         |
| Function Code |                     | 03         |
| Data          | Register address    | 7E         |
|               |                     | 32         |
|               | Number of registers | 00         |
|               |                     | 02         |

Normal response from a slave node

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 07         |
|               | Logical device ID   | 00         |
| Function Code |                     | 03         |
| Data          | Byte count          | 04         |
|               | Register data       | 00         |
|               |                     | 00         |
|               |                     | 00         |
|               |                     | 01         |

Abnormal response from a slave node

| Description |                     | Frame Data |
|-------------|---------------------|------------|
| MBAP Header | Protocol identifier | 00         |

| Description   |                   | Frame Data |
|---------------|-------------------|------------|
|               |                   | 01         |
|               | Protocol          | 00         |
|               |                   | 00         |
|               | Data length       | 00         |
|               |                   | 03         |
|               | Logical device ID | 00         |
| Function Code |                   | 83         |
| Data          | Error code        | 03         |

## 4.3.4 Writing a Single Register (0x06)

### 4.3.4.1 Frame Format of a Master Node Request

| Data Field       | Length  | Description   |
|------------------|---------|---------------|
| Function Code    | 1 byte  | 0x06          |
| Register Address | 2 bytes | 0x0000–0xFFFF |
| Register Value   | 2 bytes | 0x0000–0xFFFF |

### 4.3.4.2 Frame Format of a Normal Response from a Slave Node

| Data Field       | Length  | Description   |
|------------------|---------|---------------|
| Function Code    | 1 byte  | 0x06          |
| Register Address | 2 bytes | 0x0000–0xFFFF |
| Register Value   | 2 bytes | 0x0000–0xFFFF |

### 4.3.4.3 Frame Format of an Abnormal Response from a Slave Node

| Data Field     | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Function Code  | 1 byte | 0x86                                       |
| Exception Code | 1 byte | See <a href="#">4.3.2 Exception Code</a> . |

4.3.4.4 Examples

A master node sends a command (register address: 40200/0X9D08) to a slave node (address: 00).

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 06         |
|               | Logical device ID   | 00         |
| Function Code |                     | 06         |
| Data          | Register address    | 9D         |
|               |                     | 08         |
|               | Register data       | 00         |
|               |                     | 00         |

Normal response from a slave node

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 06         |
|               | Logical device ID   | 00         |
| Function Code |                     | 06         |
| Data          | Register address    | 9D         |
|               |                     | 08         |
|               | Register data       | 00         |
|               |                     | 00         |

Abnormal response from a slave node

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 03         |
|               | Logical device ID   | 00         |
| Function Code |                     | 86         |
| Data          | Error code          | 04         |

## 4.3.5 Writing Multiple Registers (0x10)

### 4.3.5.1 Frame Format of a Master Node Request

| Data Field                | Length             | Description   |
|---------------------------|--------------------|---------------|
| Function Code             | 1 byte             | 0x10          |
| Starting Register Address | 2 bytes            | 0x0000–0xFFFF |
| Number of Registers       | 2 bytes            | 0x0000–0x007b |
| Byte Count                | 1 byte             | 2 x <i>N</i>  |
| Register Value            | 2 x <i>N</i> bytes | Value         |

#### NOTE

*N* refers to the number of registers.

### 4.3.5.2 Frame Format of a Normal Response from a Slave Node

| Data Field       | Length  | Description   |
|------------------|---------|---------------|
| Function Code    | 1 byte  | 0x10          |
| Register Address | 2 bytes | 0x0000–0xFFFF |

| Data Field          | Length  | Description   |
|---------------------|---------|---------------|
| Number of Registers | 2 bytes | 0x0000–0x007b |

4.3.5.3 Frame Format of an Abnormal Response from a Slave Node

| Data Field     | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Function Code  | 1 byte | 0x90                                       |
| Exception Code | 1 byte | See <a href="#">4.3.2 Exception Code</a> . |

4.3.5.4 Examples

The master node sets the register address 40118/0X9CB6 to 2 and the register address 40119/0X9CB7 to 50 for the slave node (address: 00). The request frame format is as follows.

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 0B         |
|               | Logical device ID   | 00         |
| Function Code |                     | 10         |
| Data          | Register address    | 9C         |
|               |                     | B6         |
|               | Number of registers | 00         |
|               |                     | 02         |
|               | Byte count          | 04         |
|               | Register data       | 00         |
|               |                     | 02         |
|               |                     | 00         |
|               |                     | 32         |

Normal response from a slave node

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 06         |
|               | Logical device ID   | 00         |
| Function Code |                     | 10         |
| Data          | Register address    | 9C         |
|               |                     | B6         |
|               | Number of registers | 00         |
|               |                     | 02         |

Abnormal response from a slave node

| Description   |                     | Frame Data |
|---------------|---------------------|------------|
| MBAP Header   | Protocol identifier | 00         |
|               |                     | 01         |
|               | Protocol            | 00         |
|               |                     | 00         |
|               | Data length         | 00         |
|               |                     | 03         |
|               | Logical device ID   | 00         |
| Function Code |                     | 90         |
| Data          | Error code          | 04         |

4.3.6 Reading Device Identifiers (0x2B)

This command code allows reading identifiers and added packets that are relevant to the physical and function description of the remote devices.



Simulate the interface of the read device identifier as an address space. This address space consists of a set of addressable data elements. Data elements are objects to be read, and the object IDs determine these data elements.

A data element consists of three objects:

- 1. Basic device identifier: All objects of this type are mandatory, such as the vendor name, product code, and revision version.
- 2. Regular device identifier: In addition to the basic data objects, the device provides additional and optional identifiers and data object description. Define all types of objects according to definitions in the standard, but the execution of this type of objects is optional.
- 3. Extended device identifier: In addition to regular data objects, the device provides additional and optional identifiers and private data object description. All the data is related to the device.

Table 4-4 Read device identifiers

| Object ID | Object Name/Description | Type         | Mandatory/Optional | Type     |
|-----------|-------------------------|--------------|--------------------|----------|
| 0x00      | Vendor                  | ASCII string | Mandatory          | Basic    |
| 0x01      | Product code            | ASCII string | Mandatory          |          |
| 0x02      | Main revision version   | ASCII string | Mandatory          |          |
| 0x03–0x7F | --                      | --           | --                 | Regular  |
| 0x80–0xFF | --                      | --           | --                 | Extended |

4.3.6.1 Command for Querying Device Identifiers

Table 4-5 Request frame format

| Data Field     | Length | Description |
|----------------|--------|-------------|
| Function Code  | 1 byte | 0x2B        |
| MEI Type       | 1 byte | 0x0E        |
| ReadDevId Code | 1 byte | 01          |
| Object ID      | 1 byte | 0x00        |

Table 4-6 Frame format for a normal response

| Data Field       |              |               | Length  | Description |
|------------------|--------------|---------------|---------|-------------|
| Function Code    |              |               | 1 byte  | 0x2B        |
| MEI Type         |              |               | 1 byte  | 0x0E        |
| ReadDevId Code   |              |               | 1 byte  | 01          |
| Conformity Level |              |               | 1 byte  | 01          |
| More             |              |               | 1 byte  | --          |
| Next Object ID   |              |               | 1 byte  | --          |
| Object Quantity  |              |               | 1 byte  | --          |
| Object List      | First object | Object ID     | 1 byte  | 0x00        |
|                  |              | Object length | 1 byte  | N           |
|                  |              | Object value  | N bytes | --          |
|                  | ...          | ...           | ...     | ...         |

Table 4-7 Object list

| Object ID | Object Name/Description | Description                    | Type  |
|-----------|-------------------------|--------------------------------|-------|
| 0x00      | Vendor                  | "HUAWEI"                       | Basic |
| 0x01      | Product code            | "SUN2000" or "LUNA2000-P"      |       |
| 0x02      | Main revision version   | ASCII string, software version |       |

Table 4-8 Frame format for an abnormal response

| Data Field     | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Function Code  | 1 byte | 0xAB                                       |
| Exception Code | 1 byte | See <a href="#">4.3.2 Exception Code</a> . |

4.3.6.2 Command for Querying a Device List

Table 4-9 Request frame format

| Data Field     | Length | Description |
|----------------|--------|-------------|
| Function Code  | 1 byte | 0x2B        |
| MEI Type       | 1 byte | 0x0E        |
| ReadDevId Code | 1 byte | 03          |
| Object ID      | 1 byte | 0x87        |

Table 4-10 Frame format for a normal response

| Data Field       |              |               | Length  | Description |
|------------------|--------------|---------------|---------|-------------|
| Function Code    |              |               | 1 byte  | 0x2B        |
| MEI Type         |              |               | 1 byte  | 0x0E        |
| ReadDevId Code   |              |               | 1 byte  | 03          |
| Conformity Level |              |               | 1 byte  | 03          |
| More             |              |               | 1 byte  | --          |
| Next Object ID   |              |               | 1 byte  | --          |
| Object Quantity  |              |               | 1 byte  | --          |
| Object List      | First object | Object ID     | 1 byte  | 0x87        |
|                  |              | Object length | 1 byte  | N           |
|                  |              | Object value  | N bytes | --          |
|                  | ...          | ...           | ...     | ...         |

Table 4-11 Object list

| Object ID | Object Name | Type | Description                               |
|-----------|-------------|------|-------------------------------------------|
| 0x80–0x86 | Retained    | --   | Returns a null object with a length of 0. |

| Object ID | Object Name                         | Type                                                    | Description                                                                                                      |
|-----------|-------------------------------------|---------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 0x87      | Number of devices                   | int                                                     | Returns the number of devices connected to this address.                                                         |
| 0x88      | Description about the first device  | ASCII string<br>See the device description definitions. | Returns only description about the first device if an NE allows only one device to be connected to each address. |
| 0x8A      | Description about the second device | --                                                      | --                                                                                                               |
| --        | --                                  | --                                                      | --                                                                                                               |
| 0xFF      | Description about the 120th device  | --                                                      | --                                                                                                               |

4.3.6.3 Device Description Definition

Each device description consists of all "attribute=value" character strings.

"Attribute ID=%s;attribute ID=%s;... attribute ID=%s"

Example: "1= LUNA2000-200KTL-H0; 2=V800R021C10; 3=P1.0-D5.0; 4=123456789ABC; 5=1; 6=1.0; 8=LUNA2000-P"

Table 4-12 Definition of attributes

| Attribute ID | Attribute                  | Type         | Description                                                                                            |
|--------------|----------------------------|--------------|--------------------------------------------------------------------------------------------------------|
| 1            | Device model               | ASCII string | --                                                                                                     |
| 2            | Device software version    | ASCII string | --                                                                                                     |
| 3            | Interface protocol version | ASCII string | See the interface protocol version definitions.                                                        |
| 4            | ESN                        | ASCII string | --                                                                                                     |
| 5            | Device ID                  | int          | 0, 1, 2, 3,... (assigned by NEs; 0 indicates the master device into which the Modbus card is inserted) |
| 6            | Feature version            | ASCII string | --                                                                                                     |

| Attribute ID | Attribute   | Type         | Description           |
|--------------|-------------|--------------|-----------------------|
| 8            | Device type | ASCII string | LUNA2000-P or SUN2000 |

Table 4-13 Frame format for an abnormal response

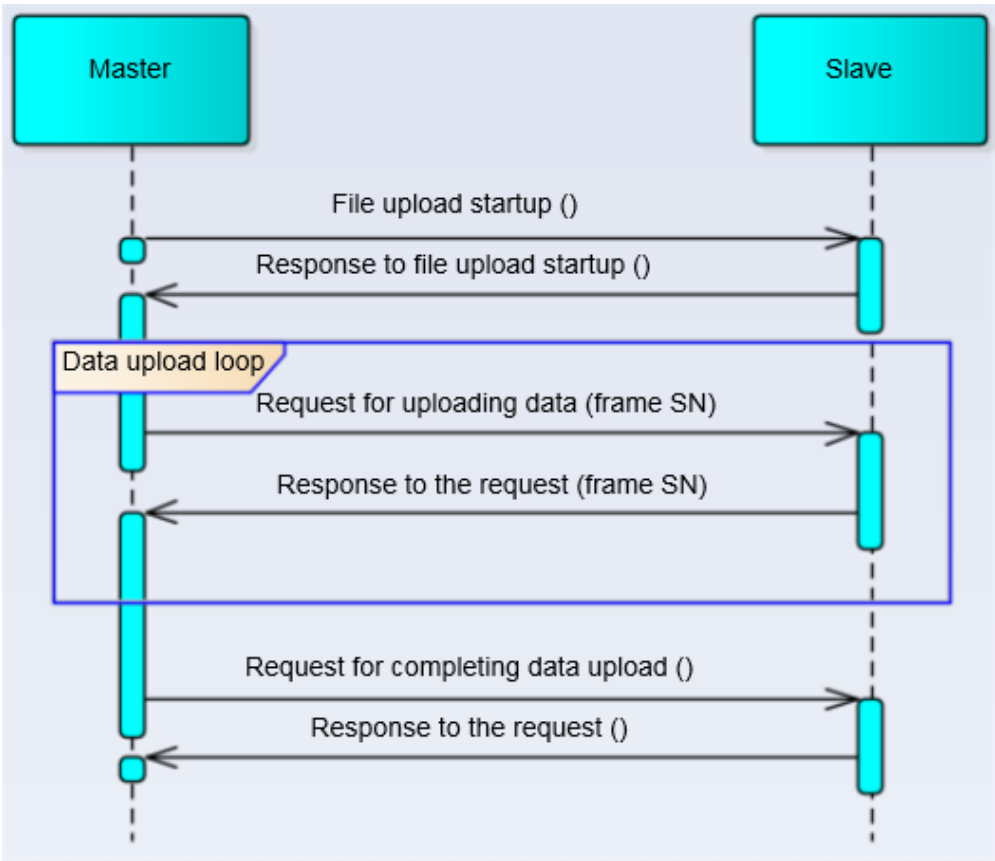
| Data Field     | Length | Description               |
|----------------|--------|---------------------------|
| Function Code  | 1 byte | 0xAB                      |
| Exception Code | 1 byte | See 4.3.2 Exception Code. |

### 4.3.7 Huawei-defined Functions (0x41)

#### 4.3.7.1 Uploading Files

Uploading files means uploading from a slave node to a master node through stream access. The following figure shows the file uploading process.

Figure 4-6 File uploading process



4.3.7.1.1 Starting the Upload

Frame format of a request from a master node

Table 4-14 PDU data field of the request frame for starting upload (0x05)

| PDU Data Field    | Length (Byte) | Description         |
|-------------------|---------------|---------------------|
| Function Code     | 1             | 0x41                |
| Sub-function Code | 1             | 0x05                |
| Data Length       | 1             | 1+N                 |
| File Type         | 1             | Unique ID of a file |
| Customized Data   | N             | -                   |

Table 4-15 PDU data field of the response frame for starting the upload (0x05)

| Data Field        | Length (Byte) | Description         |
|-------------------|---------------|---------------------|
| Function Code     | 1             | 0x41                |
| Sub-function Code | 1             | 0x05                |
| Data Length       | 1             | 6 + N               |
| File Type         | 1             | Unique ID of a file |
| File Length       | 4             | -                   |
| Data Frame Length | 1             | -                   |
| Customized Data   | N             | -                   |

**Table 4-16** PDU data field in the abnormal response frame of the slave node

| PDU Data Field | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Error Code     | 1      | 0xC1                                       |
| Exception Code | 1      | See <a href="#">4.3.2 Exception Code</a> . |

 **NOTE**

If the exception code is 0x06, the request will be retransmitted in 10 seconds. A maximum of six attempts are supported.

**4.3.7.1.2 Uploading Data**

**Table 4-17** Request frame for uploading data (0x06)

| PDU Data Field    | Length (Byte) | Description         |
|-------------------|---------------|---------------------|
| Function Code     | 1             | 0x41                |
| Sub-function Code | 1             | 0x06                |
| Data Length       | 1             | 3                   |
| File Type         | 1             | Unique ID of a file |
| Frame No.         | 2             | 0x0000–0xFFFF       |

**Table 4-18** Response frame for uploading data (0x06)

| PDU Data Field    | Length (Byte) | Description   |
|-------------------|---------------|---------------|
| Function Code     | 1             | 0x41          |
| Sub-function Code | 1             | 0x06          |
| Data Length       | 1             | 3 + N         |
| File Type         | 1             | -             |
| Frame No.         | 2             | 0x0000–0xFFFF |
| Frame Data        | N             | -             |

**Table 4-19** Abnormal response frame for uploading data

| PDU Data Field | Length (Byte) | Description                                |
|----------------|---------------|--------------------------------------------|
| Error Code     | 1             | 0xC1                                       |
| Exception Code | 1             | See <a href="#">4.3.2 Exception Code</a> . |

4.3.7.1.3 Completing the Data Upload

**Table 4-20** Request frame for completing the data upload

| PDU Data Field    | Length (Byte) | Description |
|-------------------|---------------|-------------|
| Function Code     | 1             | 0x41        |
| Sub-function Code | 1             | 0x0c        |
| Data Length       | 1             | 1           |
| File Type         | 1             | -           |

**Table 4-21** Response frame for completing the data upload

| PDU Data Field    | Length (Byte) | Description |
|-------------------|---------------|-------------|
| Function Code     | 1             | 0x41        |
| Sub-function Code | 1             | 0x0c        |
| Data Length       | 1             | 3           |
| File Type         | 1             | -           |
| File CRC          | 2             | -           |

**Table 4-22** Abnormal response frame for completing the data upload

| Data Field     | Length | Description                                |
|----------------|--------|--------------------------------------------|
| Error Code     | 1      | 0xC1                                       |
| Exception Code | 1      | See <a href="#">4.3.2 Exception Code</a> . |



4.3.7.1.4 Timeout Processing

Table 4-23 Processing specifications for sub-process timeout

| Item                                                    | Constraint |
|---------------------------------------------------------|------------|
| Response timeout period for starting an upload          | 10s        |
| Response timeout period for uploading data              | 10s        |
| Maximum retransmission attempts for data upload command | 6          |
| Response timeout period for completing a data upload    | 10s        |